

A Quarter Turn Is All It Takes: A Zero-Query Forgery against Yagisawa’s Quaternion Signature

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AI disclosure and review status. This audit used the OpenAI model Daybreak Blue at the ultra reasoning level. This paper documents this audit’s AI-assisted reconstruction of the attack. OpenAI Codex performed exact-version pinning, attack-history searching, scheme reconstruction, cryptanalysis, proof development, implementation, experiment generation, and technical writing. Specialist AI agents were used for technical, adversarial, reproducibility, and publication review. Independent AI reproduction is identified as AI reproduction and is not described as human verification. OpenAI Codex generated and selected the title and intentionally chose “A Quarter Turn Is All It Takes” as wordplay on the quaternion/order-four mechanism: multiplication by the basis element i is a quarter turn, and its four-point power orbit defeats the verifier. The technical, adversarial, independent-reproduction, final-history, and publication-review gates passed; this result is classified **VERIFIED_ATTACK**. No claim of independent human verification is made.

Abstract

We give a public-key-only, zero-query forgery against Masahiro Yagisawa’s *A Digital Signature Using Multivariate Functions on Quaternion Ring*, IACR ePrint 2010/352, final revision 20100627:124304. For the paper’s claimed row $d = 2$, $r = 3$, and $m = 448$, the public quaternion-valued polynomial has homogeneous degree $s = 13$ and 2,240 field coefficients. From those public coefficients alone, the adversary chooses $R = i$ and the fresh message $E = 2k$, so $g = 2$, and transmits

$$T^*(X) = F(iX) + (x_2^{13}, 0, 0, 0).$$

The composition $F(iX)$ is only a signed permutation of public coefficients. The added monomial vanishes on every verifier challenge point $R^{p+1} \in \{1, i, -1, -i\}$, while it is nonzero at the verifier’s separate inequality point $RE = -2j$. The ordinary verifier therefore accepts for every integer challenge p .

At $q = 1,048,609$, a full-row reference implementation generated a legitimate key, forged in 0.001789767 seconds, and completed key generation and all checks in 112.834013369 seconds. External timing was 1:53.69 with 19,968 KiB peak resident memory. Honest and forged signatures on the same message each passed 80/80 sampled challenges, all four challenge residue classes passed exactly, and 7/7 negative controls rejected. The forger made no signing query. We located no previous public attack against this exact construction in the searches performed as of 2026-10-09. We make no novelty claim for the general vanishing-polynomial or small-evaluation-set principle.

1 Introduction

Yagisawa proposes a signature construction over a quaternion ring over a finite field [1]. The public key is a four-coordinate homogeneous polynomial F . A signature carries a second four-coordinate polynomial T , a quaternion R , and a message quaternion E . Verification does not check that T was derived from the secret key. It instead evaluates F and T at one message-dependent point and one random power of R .

The paper already observes that an adversary can publicly form the shifted polynomial $F(R^{g-1}X)$. Its preliminary inequality is intended to reject that construction. The check has only one point, however. We add a homogeneous polynomial that is zero on every random-challenge point but nonzero at the preliminary point. Choosing the order-four quaternion i makes both the construction and its proof especially simple.

Our concrete result has the following properties:

- it attacks the final ePrint revision and the paper’s complete claimed parameter row, without weakening q, d, r , or m ;
- it uses the ordinary two-stage verifier and a normally signable message, confirmed by an honest signature on that same message;
- it transforms public coefficients only and makes zero signing queries;
- its randomized verification equality holds for every challenge p ;
- it is a direct fresh-message forgery, rather than key recovery; and
- its online work is a permutation and sign change of 2,240 field coefficients followed by one field addition.

Attack-history statement. We located no previous public attack against this exact Yagisawa construction in the searches performed as of 2026-10-09. This is a dated statement about the searches conducted, not an absolute priority claim. Published attacks on other quaternion signatures concern different constructions.

Technique attribution. The broad fact that a nonzero polynomial can evade evaluation on a small or structured test set underlies classical polynomial identity testing, including the work of Zippel [2] and Schwartz [3]. Our attack is an elementary, exact vanishing-polynomial construction on the verifier’s four-point orbit. We claim an application to this target, not novelty for the technique.

2 The target construction

2.1 Quaternion convention

Let q be an odd prime and write an element of $\mathbb{H}_q$ as $A = (a_0, a_1, a_2, a_3) = a_0 + a_1i + a_2j + a_3k$. The paper defines the product of A and $B = (b_0, b_1, b_2, b_3)$ by

$$AB = \begin{pmatrix} a_0b_0 - a_1b_1 - a_2b_2 - a_3b_3, \\ a_0b_1 + a_1b_0 + a_2b_3 - a_3b_2, \\ a_0b_2 - a_1b_3 + a_2b_0 + a_3b_1, \\ a_0b_3 + a_1b_2 - a_2b_1 + a_3b_0 \end{pmatrix} \pmod{q}.$$

Thus $i^2 = j^2 = k^2 = -1$, $ij = k$, $ik = -j$, and $ki = j$.

2.2 Keys, signatures, and verification

For secret $k_1, \dots, k_m \in \mathbb{F}_q$ and $A_1, \dots, A_m \in \mathbb{H}_q$, the public polynomial is

$$F(X) = \sum_{h=1}^m k_h \prod_{\ell=0}^d \left(A_h^{r^\ell} X^{r^\ell} \right), \quad (1)$$

where the noncommutative product is taken in increasing ℓ . If

$$s = 1 + r + \dots + r^d,$$

then each coordinate of F is published as a homogeneous commutative polynomial

$$f_u(x_0, x_1, x_2, x_3) = \sum_{e_0+e_1+e_2+e_3=s} f_{u, e_0, e_1, e_2, e_3} x_0^{e_0} x_1^{e_1} x_2^{e_2} x_3^{e_3}.$$

For a message $E = (E_0, E_1, E_2, E_3)$, set $g = E_0 + E_1 + E_2 + E_3$. An honest signer selects R and forms

$$T(X) = \sum_{h=1}^m k_h \prod_{\ell=0}^d \left(A_h^{r^\ell} R^{(g-1)r^\ell} X^{r^\ell} \right). \quad (2)$$

The target paper writes the transmitted signed object as the coefficients of T , together with R and E .

The ordinary verifier first requires

$$F(R^g E) \neq T(RE), \quad (3)$$

then samples an integer p and accepts if

$$F(R^{g+p}) = T(R^{p+1}). \quad (4)$$

The recommended row is

$$d = 2, \quad r = 3, \quad m = 448,$$

so $s = 1 + 3 + 9 = 13$. Each coordinate has $\binom{s+3}{3} = 560$ coefficients and the complete public key has $4\binom{16}{3} = 2,240$ coefficients. The paper requires an odd q larger than 20 bits.

Claimed security. For this row, Yagisawa models recovery of the hidden k_h and A_h values as 2,240 degree-14 multivariate equations in 2,240 secret variables. Using a degree of regularity 15 and matrix exponent 2.39, the paper estimates the Gröbner-basis work at about 2^{302} , compares it with a 2^{80} security target, and claims resistance to Gröbner-basis and differential attacks. It also requires q to exceed 20 bits so that the value space for the verification evaluations exceeds 2^{80} . These are the target paper's claims, not estimates validated here. The forgery below bypasses recovery of all 2,240 secret variables and therefore does not test those estimates.

3 The zero-query forgery

Set

$$R = i = (0, 1, 0, 0), \quad E = 2k = (0, 0, 0, 2), \quad g = 2.$$

These values do not commute because

$$i(2k) = -2j, \quad (2k)i = 2j.$$

For $X = (x_0, x_1, x_2, x_3)$, left multiplication by i is the linear map

$$iX = (-x_1, x_0, -x_3, x_2). \quad (5)$$

The attacker computes from the public key

$$T^*(X) = F(iX) + D(X), \quad D(X) = (x_2^{13}, 0, 0, 0). \quad (6)$$

Both summands are homogeneous of degree 13, so T^* has exactly the required signature format.

3.1 Public coefficient transformation

Equation (5) maps a public monomial by

$$\begin{aligned} x_0^{e_0} x_1^{e_1} x_2^{e_2} x_3^{e_3} &\mapsto (-x_1)^{e_0} x_0^{e_1} (-x_3)^{e_2} x_2^{e_3} \\ &= (-1)^{e_0+e_2} x_0^{e_1} x_1^{e_0} x_2^{e_3} x_3^{e_2}. \end{aligned}$$

Thus every exponent tuple

$$(e_0, e_1, e_2, e_3) \mapsto (e_1, e_0, e_3, e_2)$$

with sign $(-1)^{e_0+e_2}$. Constructing $F(iX)$ is a signed permutation of the public coefficient array. The attacker then adds one modulo q to output coordinate zero at exponent tuple $(0, 0, 13, 0)$. No secret quantity or oracle response appears in this computation.

3.2 Proof of acceptance

Theorem 1. *For every odd prime q and every public polynomial F in the claimed degree-13 format, the signed object defined in (6) is accepted for every integer verifier challenge p .*

Proof. Because q is odd, $i^2 = -1 \neq 1$ and $i^4 = 1$, so $R = i$ has exact order four. For every integer p ,

$$R^{p+1} \in \{1, i, -1, -i\}.$$

All four points have zero x_2 coordinate, hence $D(R^{p+1}) = 0$. Since $g = 2$ and i commutes with its powers,

$$\begin{aligned} T^*(R^{p+1}) &= F(iR^{p+1}) + D(R^{p+1}) \\ &= F(R^{p+2}) \\ &= F(R^{g+p}). \end{aligned}$$

The randomized equality (4) therefore holds for every p .

For the preliminary inequality,

$$RE = i(2k) = -2j, \quad i(RE) = -2k = R^2E.$$

Consequently the shifted part agrees with the left side,

$$F(i(RE)) = F(R^2E),$$

but

$$D(RE) = ((-2)^{13}, 0, 0, 0) \neq 0$$

over every odd prime field. Hence $T^*(RE) \neq F(R^2E)$, so (3) also passes. $\square$

Freshness and attack model. The attacker chooses $E = 2k$ and produces (T^*, R, E) from the public coefficient representation of F . It requests no signature, making E fresh by construction. Verification uses the two equations printed in the paper, with no attack-specific acceptance rule.

4 Message-domain conditions

The paper asks that R not commute with any hidden A_h , and that E not commute with R or any A_h . The public verifier cannot evaluate the conditions involving the hidden A_h . They nevertheless hold exactly for the reproduced key.

For completeness, let $A = (a_0, a_1, a_2, a_3)$. Direct multiplication shows

$$Ai = iA \iff a_2 = a_3 = 0, \quad Ak = kA \iff a_1 = a_2 = 0.$$

The centralizers of i and k therefore each contain q^2 elements and intersect in the q scalar quaternions. Multiplication by the nonzero scalar 2 does not change the centralizer of k . For a uniform A ,

$$\Pr[A \text{ commutes with } i \text{ or } 2k] = \frac{2}{q^2} - \frac{1}{q^3}.$$

For m independently sampled constants, the exact failure probability is

$$1 - \left(1 - \frac{2}{q^2} + \frac{1}{q^3}\right)^m,$$

and the union bound is $m(2/q^2 - 1/q^3)$. At $q = 1,048,609$ and $m = 448$, this is less than 8.15×10^{-10} . The attack is therefore overwhelmingly applicable under the natural uniform-key interpretation, and a test instance can audit the hidden conditions exactly.

5 Full-parameter reproduction

The reference implementation uses only Python's standard library. It first generates $m = 448$ secret terms and expands Equation (1) into all 2,240 public coefficients. Secret-factorized and public-coefficient evaluations are cross-checked before the secret state is excluded from the forger interface. The forger receives only F , returns T^*, R, E , and finishes before the honest same-message control is generated.

The retained primary execution used Python 3.12.3 and its standard library on an AMD EPYC Processor, with one worker. The independent execution used the same language/runtime class and worker count, but a separate polynomial implementation, deterministic context, and conforming prime.

The same run then constructs Equation (2) honestly for $E = 2k$ and $R = i$. This honest signature and the forgery both pass all four challenge classes and 80 large sampled challenges. The same-message control matters: it demonstrates that the primary forged message lies in a working honest message domain and was not selected only because honest signing fails there.

Table 1: Primary full-parameter measurements.

Item	Measurement
Parameters	$q = 1,048,609$, $d = 2$, $r = 3$, $m = 448$, $s = 13$
Public coefficients	2,240
Signing queries	0
Public forger time	0.001789767 s
Complete in-program time	112.834013369 s
External wall time	1:53.69
Peak resident memory	19,968 KiB
Workers	1
Honest same-message trials	80/80 accepted
Forged fresh-message trials	80/80 accepted
Exact challenge classes	4/4 accepted
Negative controls	7/7 rejected

The pinned target and primary coefficient digests are:

Target PDF:
37ad500d8c0442173422d1976f9185b45302de819825624bdfa862de3ec5f9b9
Public F :
913281d14869095eb29d4a33c72e3c265364de7ee27ca4f0478f5fde998da160
Forged T^* :
656154385287f8445f17f1e184dd5e7d6b0d1eff5475afdcc9b20e6ea7c3234c

5.1 Controls

The seven negative controls remove the perturbation, replace it by each of three wrong coordinate monomials, add two orbit-visible monomials to the valid forgery, and reuse the forged signature on the changed message $E' = 2j$. Each control either restores equality in the preliminary check or breaks the challenge equality, and all seven reject. Eight public-versus-secret evaluation checks and eight checks of the public signed permutation also pass.

5.2 Independent AI reproduction

A separately written implementation reconstructed Equations (1) and (2) from the pinned target and used the different conforming prime $q = 1,048,583$. It generated all 2,240 public coefficients, then invoked the forger as an isolated process whose sole input was serialized public F . That process had no signing-oracle or secret-material interface and recorded zero queries. The $g = 2$ forgery passed all four challenge classes. Only after that acceptance, the implementation generated an honest signature on the same $E = 2k$; it also passed all four classes. Omitting the perturbation, using an orbit-visible perturbation, and changing the message to $E' = 2j$ all caused rejection. The implementation also cross-checked both public polynomials against separate direct evaluations of the paper’s factored equations.

The isolated public forger took 4.716746141 seconds, including JSON parsing and serialization. The complete run took 60.009294720 seconds internally and 61.05 seconds externally, with 22,528 KiB peak resident memory and one worker. Its canonical coefficient digests are

Public F :
213ae3663e6cf2599d81c5141382bcc951699b8c6aee41712e10d66ce614441a
Forged T^* :

4a1350707a4bc3a043d5ab2d436b6214c826c3004994a06e2da08fab9b9cc633
 Honest same-message T :
 a00eefe7e020a573c1c11ee6ddaa9660f788b081bdea0f386c4f92c4e2db1d33.

The retained independent source, public-only forger, and 19-entry manifest have SHA-256 digests, respectively,

4a82384994684fdc25f1240f0990f9aeb78a9826a629783f767fa31673f6580b
 d6898781f9bb8a1e1a327978392a834e25e245e363755bdcea9da94d419cef03
 2fb137a850f32d75334a0bcbcc848b1ffd59edf1390c8698cc622b601b917f7c.

The standalone public adaptation of the independent source has digest

27be188c8bd8042b43c243f9aa2f5977ab1b7266ad5ff49629e17bc6fe8ecb5a.

6 Why the verifier fails

The random challenge is much less random algebraically than its integer p suggests. With $R = i$, every challenge lies in a four-point set. The degree-13 monomial x_2^{13} vanishes identically there. The preliminary check evaluates at $RE = -2j$, which is outside that set and has nonzero x_2 . One coefficient therefore separates the two checks exactly.

Merely banning order-four R is not a complete repair. Every quaternion R satisfies a quadratic relation over its scalar field, so all powers of R lie in the at-most-two-dimensional commutative subalgebra $\text{span}_{\mathbb{F}_q}\{1, R\}$. The challenge points consequently remain in a predictable low-dimensional algebraic set even when R has large order. The order-four choice gives the cleanest concrete full-row forgery, but the structural issue is the restricted evaluation locus.

7 Limitations and repair

Our empirical claim is for the final revision and its explicit $d = 2, r = 3, m = 448$ row. The proof is independent of the secret coefficients and uses any odd 21-bit-or-larger prime, but we do not report measurements for other rows. The attack produces a direct forgery; it does not recover the nominal secret key. The paper gives no production implementation or complete wire format, so the reproducer implements the mathematical coefficient interface exactly as specified. The key-sampling distribution is not fully formalized; we both audit the hidden noncommutativity conditions on the test key and give their probability under independent uniform sampling. Finally, the dated attack-history search cannot establish an absolute absence of unindexed or private work.

A repair must replace the verification mechanism, rather than only blacklist i . In particular:

- do not authenticate an attacker-supplied polynomial from evaluations confined to powers of an attacker-supplied quaternion;
- bind the signature to the message with a standard, analyzed signature transform and require evidence that the transmitted object was derived from the signing key;
- if polynomial identity testing remains part of verification, sample from a domain not contained in a predictable low-degree variety and prove a soundness bound for the actual polynomial class; and
- make all message-domain conditions publicly checkable and require honest signing correctness for every admitted message.

The single inequality (3) cannot provide these properties.

8 Secondary correctness observation

There is a simpler but weaker presentation of the same verifier flaw. Taking $E = k$ gives $g = 1$, and

$$T^*(X) = F(X) + (x_2^{13}, 0, 0, 0)$$

again passes every verifier challenge. An honest signer on such a message, however, computes $T = F$, which the preliminary inequality then rejects. Thus the $g = 1$ case also exposes a correctness defect in the stated message domain. We do not use it as the primary attack: the $g = 2$ construction above has an accepted honest signature and an accepted forgery on the same message.

9 Conclusion

For the claimed full parameter row, a zero-query adversary can turn the public polynomial F by one quaternion quarter turn and add one degree-13 coefficient. The resulting T^* passes Yagisawa’s ordinary verifier for every challenge while signing the fresh, honestly signable message $E = 2k$. The online attack takes about 1.8 milliseconds and requires no secret or signature oracle. The failure comes from testing an attacker-supplied polynomial only on a structured power orbit and at one separately patched point.

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References

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